

PE-EC603D — Information Theory and Coding

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Module 2 Exam Notes: Markov Sources & Channel Capacity

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Markov Sources

Information sources are categorized based on whether current outputs depend on past outputs. A source is **memoryless** if symbols are emitted independently. A source has **memory** if current output probabilities are conditioned on previous symbols.

1.1 Definition of Markov Sources

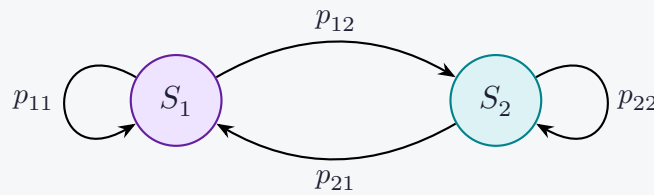
Markov Source: A discrete source with memory where the probability of emitting the next symbol depends on a finite number of preceding symbols.

In a **first-order Markov source**, the probability of the next symbol depends only on the immediately preceding symbol:

$$P(S_t = s_j \mid S_{t-1} = s_i, S_{t-2} = s_k, \dots) = P(S_t = s_j \mid S_{t-1} = s_i) = p_{ij} \quad (1)$$

Where p_{ij} is the transition probability from state s_i to state s_j .

State Transition Diagram of a 2-State Markov Source



1.2 Transition Probability Matrix (TPM)

The transition probabilities of a Markov chain with K states are organized into a square matrix P :

$$P = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1K} \\ p_{21} & p_{22} & \cdots & p_{2K} \\ \vdots & \vdots & \ddots & \vdots \\ p_{K1} & p_{K2} & \cdots & p_{KK} \end{bmatrix} \quad (2)$$

Since the source must transition to one of the possible states, the sum of probabilities in each row is exactly 1:

$$\sum_{j=1}^K p_{ij} = 1 \quad \text{for all } i = 1, 2, \dots, K \quad (3)$$

1.3 Stationary State Probabilities

Let $\mathbf{W} = [w_1, w_2, \dots, w_K]$ represent the stationary (steady-state) probability vector, where w_i is the probability that the source is in state i in the long run.

Stationary State Probability Equations

The stationary probability vector satisfies:

$$\mathbf{W}P = \mathbf{W} \quad \text{and} \quad \sum_{i=1}^K w_i = 1 \quad (4)$$

Which yields a system of linear equations:

$$w_1 p_{11} + w_2 p_{21} + \cdots + w_K p_{K1} = w_1 \quad (5)$$

$$w_1 p_{12} + w_2 p_{22} + \cdots + w_K p_{K2} = w_2 \quad (6)$$

$$\vdots \quad (7)$$

$$w_1 + w_2 + \cdots + w_K = 1 \quad (8)$$

1.4 Entropy of Markov Sources (Entropy Rate)

Because the output probability depends on the current state, we first calculate the entropy of each state individually, and then compute the weighted average over all states.

Markov Source Entropy Rate Formulas

- Entropy of State i (H_i):** The uncertainty of the next symbol given the source is currently in state i .

$$H_i = - \sum_{j=1}^K p_{ij} \log_2 p_{ij} \quad (\text{bits/symbol}) \quad (9)$$

- Average Entropy of Markov Source ($H(X)$):** The long-run average uncertainty per symbol.

$$H(X) = \sum_{i=1}^K w_i H_i = - \sum_{i=1}^K w_i \sum_{j=1}^K p_{ij} \log_2 p_{ij} \quad (\text{bits/symbol}) \quad (10)$$

Worked Example: Entropy Rate of a Markov Source

Problem: A 2-state Markov source has the following transition probability matrix:

$$P = \begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{bmatrix}$$

- Find the stationary state probabilities w_1 and w_2 .
- Calculate the state entropies H_1 and H_2 .
- Find the average entropy rate $H(X)$ of this source.

Solution:

(a) Find stationary probabilities w_1 and w_2 : Using $\mathbf{W}P = \mathbf{W}$:

$$[w_1, w_2] \begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{bmatrix} = [w_1, w_2]$$

This yields the equations:

$$\begin{aligned} 0.8w_1 + 0.3w_2 &= w_1 \implies 0.3w_2 = 0.2w_1 \implies w_2 = \frac{2}{3}w_1 \\ 0.2w_1 + 0.7w_2 &= w_2 \implies 0.2w_1 = 0.3w_2 \quad (\text{redundant}) \end{aligned}$$

Using the probability normalization constraint $w_1 + w_2 = 1$:

$$w_1 + \frac{2}{3}w_1 = 1 \implies \frac{5}{3}w_1 = 1 \implies w_1 = 0.6 \quad \text{and} \quad w_2 = 0.4$$

(b) Calculate state entropies H_1 and H_2 :

$$\begin{aligned} H_1 &= -[0.8 \log_2 0.8 + 0.2 \log_2 0.2] = -[0.8(-0.3219) + 0.2(-2.3219)] = 0.7219 \text{ bits/symbol} \\ H_2 &= -[0.3 \log_2 0.3 + 0.7 \log_2 0.7] = -[0.3(-1.7370) + 0.7(-0.5146)] = 0.8813 \text{ bits/symbol} \end{aligned}$$

(c) Find the average entropy rate $H(X)$:

$$\begin{aligned} H(X) &= w_1 H_1 + w_2 H_2 \\ &= (0.6 \times 0.7219) + (0.4 \times 0.8813) \\ &= 0.4331 + 0.3525 = 0.7856 \text{ bits/symbol} \end{aligned}$$

Answer: $w_1 = 0.6, w_2 = 0.4; H_1 = 0.7219, H_2 = 0.8813$ bits/sym; average entropy is **0.7856 bits/symbol**.

2. Shannon's Second Theorem (Noisy Channel Coding Theorem)

When transmitting data over physical media, noise corrupts signals, causing bit errors. Shannon proved that error-free communication is possible even over noisy channels, provided the transmission rate does not exceed a threshold called **Channel Capacity**.

2.1 Channel Capacity Concept

Channel Capacity (C): The maximum rate at which information can be transmitted reliably through a communication channel. It is mathematically defined as the maximum of the mutual information $I(X; Y)$ over all possible input probability distributions $P(x)$:

$$C = \max_{P(x)} I(X; Y) \quad (\text{bits/channel-use}) \quad (11)$$

2.2 Noisy Channel Coding Theorem

Shannon's Second Theorem (Noisy Channel Coding Theorem)

Let a discrete memoryless channel have capacity C .

- Theorem:** If the source transmission rate R (bits/second) is less than or equal to C ($R \leq C$), there exists a coding scheme (with block length N sufficiently large) such

Frequently Asked / Viva Questions (contd.)

that the probability of error at the receiver, P_e , can be made arbitrarily small:

$$\lim_{N \rightarrow \infty} P_e = 0 \quad (12)$$

2. **Converse:** If $R > C$, it is impossible to design a coding scheme that achieves error-free transmission. The probability of bit error P_e is bounded away from zero, and reliable communication cannot occur.

This theorem is historically significant because it proved that channel noise limits the **rate of transmission**, not its **accuracy**. We do not need noiseless channels for error-free data transfer; we only need appropriate error-correcting codes.

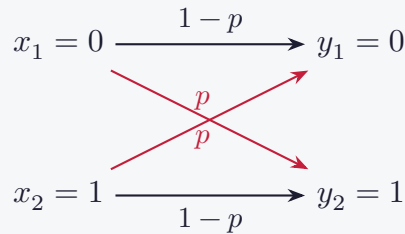
3. Calculation of Channel Capacity for Discrete Channels

A discrete channel is represented by input alphabet X , output alphabet Y , and a channel transition matrix containing conditional probabilities $P(y_j|x_i)$.

3.1 Binary Symmetric Channel (BSC)

The BSC is the most common model for binary transmission. The input and output alphabets are binary $\{0, 1\}$. The probability of a bit being transmitted correctly is $1 - p$, and the probability of a bit being flipped (error) is p .

Binary Symmetric Channel (BSC) Transition Model



3.1.1 Derivation of BSC Channel Capacity

We calculate capacity using $I(X; Y) = H(Y) - H(Y|X)$. First, evaluate the conditional entropy $H(Y|X)$:

$$\begin{aligned} H(Y|X) &= - \sum_{i=1}^2 P(x_i) \sum_{j=1}^2 P(y_j|x_i) \log_2 P(y_j|x_i) \\ &= -P(x_1) [P(y_1|x_1) \log_2 P(y_1|x_1) + P(y_2|x_1) \log_2 P(y_2|x_1)] \\ &\quad - P(x_2) [P(y_1|x_2) \log_2 P(y_1|x_2) + P(y_2|x_2) \log_2 P(y_2|x_2)] \\ &= -P(x_1) [(1-p) \log_2(1-p) + p \log_2 p] - P(x_2) [p \log_2 p + (1-p) \log_2(1-p)] \end{aligned}$$

Since $P(x_1) + P(x_2) = 1$:

$$H(Y|X) = -p \log_2 p - (1-p) \log_2 (1-p) = H(p) \quad (\text{independent of input distribution } P(x))$$

Substitute this into the capacity equation:

$$C = \max_{P(x)} [H(Y) - H(Y|X)] = \max_{P(x)} H(Y) - H(p)$$

Since Y is a binary variable, its entropy $H(Y)$ is maximized when Y has equiprobable outputs, i.e., $P(y_1) = P(y_2) = 0.5$, which yields $H(Y)_{\max} = 1$. This maximum is achieved when the input symbols are equiprobable ($P(x_1) = P(x_2) = 0.5$).

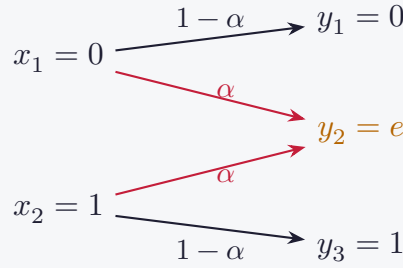
BSC Channel Capacity

$$C_{\text{BSC}} = 1 - H(p) = 1 + p \log_2 p + (1-p) \log_2 (1-p) \quad (\text{bits/channel-use}) \quad (13)$$

3.2 Binary Erasure Channel (BEC)

In a BEC, the transmitter sends binary digits $\{0, 1\}$. Due to noise, the receiver may fail to decode the bit. Rather than receiving an incorrect bit, the receiver recognizes a lost bit and labels it as an erasure symbol ' e '. The probability of erasure is α .

Binary Erasure Channel (BEC) Transition Model



3.2.1 Capacity of the BEC

Let the input probability be $P(x = 0) = \beta$, and $P(x = 1) = 1 - \beta$. The output probabilities are:

$$\begin{aligned} P(y = 0) &= (1 - \alpha)\beta \\ P(y = e) &= \alpha\beta + \alpha(1 - \beta) = \alpha \\ P(y = 1) &= (1 - \alpha)(1 - \beta) \end{aligned}$$

To find capacity, we compute conditional entropy $H(Y|X)$: Since for any input x , the output is correct with probability $1 - \alpha$ and erased with probability α :

$$H(Y|X) = H(\alpha) = -\alpha \log_2 \alpha - (1 - \alpha) \log_2 (1 - \alpha)$$

The output entropy $H(Y)$ is:

$$\begin{aligned} H(Y) &= -P(y = 0) \log_2 P(y = 0) - P(y = e) \log_2 P(y = e) - P(y = 1) \log_2 P(y = 1) \\ &= -(1 - \alpha)\beta \log_2 [(1 - \alpha)\beta] - \alpha \log_2 \alpha - (1 - \alpha)(1 - \beta) \log_2 [(1 - \alpha)(1 - \beta)] \\ &= -\alpha \log_2 \alpha - (1 - \alpha) \log_2 (1 - \alpha) + (1 - \alpha) [-\beta \log_2 \beta - (1 - \beta) \log_2 (1 - \beta)] \\ &= H(\alpha) + (1 - \alpha)H(X) \end{aligned}$$

Using $I(X; Y) = H(Y) - H(Y|X)$:

$$I(X; Y) = H(\alpha) + (1 - \alpha)H(X) - H(\alpha) = (1 - \alpha)H(X)$$

To maximize $I(X; Y)$, we maximize $H(X)$. Since X is binary, its maximum entropy is $H(X)_{\max} = 1$ bit (at $\beta = 0.5$).

BEC Channel Capacity

$$C_{\text{BEC}} = 1 - \alpha \quad (\text{bits/channel-use}) \quad (14)$$

This shows that erasures reduce capacity linearly: if 20% of bits are erased ($\alpha = 0.2$), the capacity is 80% of a noiseless channel ($C = 0.8$ bits/use).

4. Application to Continuous Channels

Discrete channel models fail when signals are continuous voltage waveforms corrupted by additive white Gaussian noise (AWGN). We analyze these systems using continuous capacities.

4.1 The Shannon-Hartley Theorem

Shannon-Hartley Theorem: The maximum rate of transmission over a continuous channel of bandwidth B (Hz), corrupted by Additive White Gaussian Noise (AWGN) with signal power S and noise power N , is:

$$C = B \log_2 \left(1 + \frac{S}{N} \right) \quad (\text{bits/second}) \quad (15)$$

If noise spectral density is $N_0/2$ (double-sided) or N_0 (single-sided), the total noise power is $N = N_0 B$:

$$C = B \log_2 \left(1 + \frac{S}{N_0 B} \right) \quad (16)$$

4.2 Implications of Shannon-Hartley Theorem

This relation highlights several trade-offs:

- Signal-to-Noise Ratio (SNR) vs Bandwidth (B):** We can exchange bandwidth for SNR. A system with small bandwidth can achieve high capacity if the SNR is high, and vice-versa.
- Infinite Bandwidth capacity limit:** If we keep signal power S and noise density N_0 constant, and let the bandwidth B grow to infinity, the capacity does not grow to infinity. Instead, it reaches a thermodynamic limit:

4.2.1 Derivation of Infinite Bandwidth Capacity

We evaluate the limit of C as $B \rightarrow \infty$:

$$\lim_{B \rightarrow \infty} C = \lim_{B \rightarrow \infty} B \log_2 \left(1 + \frac{S}{N_0 B} \right)$$

Let $x = \frac{S}{N_0 B}$. As $B \rightarrow \infty$, $x \rightarrow 0$. We express B as $B = \frac{S}{N_0 x}$:

$$\begin{aligned} \lim_{B \rightarrow \infty} C &= \lim_{x \rightarrow 0} \frac{S}{N_0 x} \log_2(1 + x) \\ &= \frac{S}{N_0} \lim_{x \rightarrow 0} \log_2(1 + x)^{1/x} \end{aligned}$$

By the definition of the natural base e , $\lim_{x \rightarrow 0} (1 + x)^{1/x} = e$:

$$\lim_{B \rightarrow \infty} C = \frac{S}{N_0} \log_2 e \approx 1.4427 \frac{S}{N_0}$$

Infinite Bandwidth Capacity Limit

$$C_\infty = 1.44 \frac{S}{N_0} \quad (\text{bits/second}) \quad (17)$$

This represents the absolute physical limit on transmission capacity under a power constraint, even with infinite bandwidth.

4.3 Bandwidth-Efficiency Trade-off (C/B vs. E_b/N_0)

We define the **spectral efficiency** as C/B (bps/Hz). Let E_b be the energy per bit transmitted. Since signal power is energy per unit time:

$$S = C \times E_b$$

Substitute this into the Shannon-Hartley equation:

$$\begin{aligned} \frac{C}{B} &= \log_2 \left(1 + \frac{C E_b}{B N_0} \right) \\ 2^{C/B} &= 1 + \frac{C}{B} \left(\frac{E_b}{N_0} \right) \end{aligned}$$

Solving for E_b/N_0 :

Bandwidth-Efficiency Relation

$$\frac{E_b}{N_0} = \frac{2^{C/B} - 1}{C/B} \quad (18)$$

4.3.1 Derivation of the Shannon Limit

To find the absolute minimum value of E_b/N_0 required for error-free communication, we take the limit as the spectral efficiency $C/B \rightarrow 0$ (highly redundant coding over very wide bandwidth):

$$\lim_{C/B \rightarrow 0} \frac{E_b}{N_0} = \lim_{y \rightarrow 0} \frac{2^y - 1}{y}$$

Using L'Hôpital's rule (differentiating numerator and denominator with respect to y):

$$\lim_{y \rightarrow 0} \frac{\ln 2 \cdot 2^y}{1} = \ln 2 \approx 0.693$$

Expressed in decibels (dB):

$$\left(\frac{E_b}{N_0} \right)_{\text{dB}} = 10 \log_{10}(0.693) \approx -1.59 \text{ dB}$$

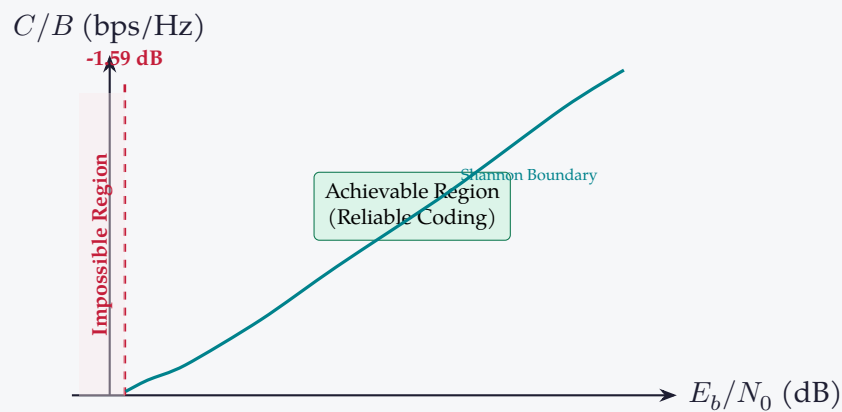
The Shannon Limit

The minimum energy-to-noise ratio required for reliable transmission is:

$$\left(\frac{E_b}{N_0} \right)_{\text{min}} = -1.59 \text{ dB} \quad (19)$$

If a system operates below -1.59 dB , error-free communication is physically impossible.

Spectral Efficiency C/B vs E_b/N_0 (Shannon Boundary)



Quick Revision — Module II

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Key Formulas at a Glance

Markov Transition Prob.	$\sum_{j=1}^K p_{ij} = 1$ (Row sum of TPM equals 1)
Steady-State Vector	$\mathbf{W}P = \mathbf{W}$ and $\sum w_i = 1$
State Entropy	$H_i = -\sum_{j=1}^K p_{ij} \log_2 p_{ij}$
Markov Source Entropy	$H(X) = \sum w_i H_i$
Channel Capacity	$C = \max_{P(x)} I(X; Y)$ (bits/channel-use)
BSC Capacity	$C_{\text{BSC}} = 1 - H(p) = 1 + p \log_2 p + (1 - p) \log_2 (1 - p)$
BEC Capacity	$C_{\text{BEC}} = 1 - \alpha$
Shannon-Hartley Theorem	$C = B \log_2 \left(1 + \frac{S}{N}\right) = B \log_2 \left(1 + \frac{S}{N_0 B}\right)$ (bps)
Infinite Bandwidth Limit	$C_\infty = 1.44 \frac{S}{N_0}$
Shannon Limit (Min SNR)	$\left(\frac{E_b}{N_0}\right)_{\min} = \ln 2 \approx 0.693 \approx -1.59 \text{ dB}$

One-Line Definitions

- **Markov Source:** A source with memory where current output statistics depend on previous state history.
- **Transition Matrix (TPM):** A matrix storing conditional probabilities of moving between Markov states.
- **Steady-State Vector:** Stationary probabilities indicating long-run occupancy rates of Markov states.
- **Channel Capacity:** The maximum rate of information transmission that can be achieved over a channel.
- **Binary Symmetric Channel:** A binary channel model with symmetric error (crossover) probability p .
- **Binary Erasure Channel:** A channel model where bits are either received correctly or recognized as lost (erased).
- **AWGN Channel:** A continuous channel model corrupted by additive white Gaussian noise.
- **Shannon Limit:** The absolute minimum value of E_b/N_0 (-1.59 dB) required for error-free transmission.

Frequently Asked / Viva Questions

Q1. State the main condition that defines a first-order Markov source.

The output probability at any time instant depends only on the output symbol of the immediately preceding time instant.

Q2. Why does the sum of transition probabilities in each row of a TPM equal 1?

A row represents all transition possibilities leaving a specific state. The source must transition to one of the possible states, so the sum of these probabilities is certain (1).

Q3. Explain the physical meaning of Shannon's Second Theorem.

If transmission rate $R \leq C$, error-free communication is possible using coding. If $R > C$, it is physically impossible to achieve error-free communication.

Q4. What happens to the capacity of a BSC when the crossover probability $p = 0.5$?

Capacity $C = 1 - H(0.5) = 1 - 1 = 0$. The channel output is completely random and independent of input, meaning no information can be transmitted.

Q5. What is an "erasure" symbol in the context of a BEC?

It is a symbol received as a lost bit (labeled 'e'), meaning the receiver knows a bit was sent but cannot determine if it was '0' or '1'.

Q6. What are the components of noise in the Shannon-Hartley theorem?

Additive (noise is added to signal), White (flat power spectral density across all frequencies), and Gaussian (probability density function of noise voltage is Gaussian).

Q7. Why can't we achieve infinite channel capacity by making the bandwidth infinite?

As bandwidth increases, the noise power $N = N_0 B$ also increases. The capacity reaches a thermodynamic limit of $1.44S/N_0$ bps.

Q8. Explain the bandwidth-SNR trade-off using a practical example.

In telephone channels with small bandwidth, high SNR is used to achieve high rates. In satellite links with low power (low SNR), wide bandwidths are used.

Q9. State the value of the Shannon Limit and its practical significance.

-1.59 dB. It is the absolute minimum signal-to-noise ratio per bit required to transmit information over an AWGN channel without errors.

Q10. Under what conditions does the capacity of a channel equal its bandwidth?

When the signal-to-noise ratio $S/N = 1$ (or 0 dB), because $C = B \log_2(1 + 1) = B \log_2(2) = B$ bps.

Q11. Distinguish between a lossless channel and a noiseless channel.

In a lossless channel, $H(Y|X) = 0$ (no noise, inputs map uniquely to outputs). In a noiseless channel, $H(X|Y) = 0$ (outputs map uniquely to inputs, no ambiguity).

Frequently Asked / Viva Questions (contd.)

Q12. How do you calculate the entropy of a Markov source?

First, find steady-state probabilities $\mathbf{W}$. Then, calculate state entropies H_i . The average source entropy is the weighted sum $H(X) = \sum w_i H_i$.

Comparison of BSC and BEC Models

Parameter	Binary Symmetric Channel (BSC)	Binary Erasure Channel (BEC)
Input Alphabet	Binary: $\{0, 1\}$	Binary: $\{0, 1\}$
Output Alphabet	Binary: $\{0, 1\}$	Ternary: $\{0, e, 1\}$ (where e is erasure)
Error Type	Bit flip (crossover), $0 \leftrightarrow 1$.	Bit loss, input becomes unreadable symbol e .
Error Knowledge	Receiver does not know which bits are flipped.	Receiver knows exactly which bits are erased.
Capacity Formula	$C = 1 - H(p)$ bits/use	$C = 1 - \alpha$ bits/use